

SEARCH QUERIES

Database Search Strings and Retrieval Results

Search Strategy

Search syntax was adapted to the indexing conventions of each database while preserving semantic equivalence. Retrieved records were exported in compatible bibliographic formats.

The complete Search Query for the ACM Library is available in Appendix 1 of the article.

IEEE Xplore

Search Field	Full Text & Metadata
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IEEE Xplore requires explicit field specification. Searches were therefore conducted using the Full Text & Metadata field.

(1) CT Evaluation / Framework Query

```
("Full Text & Metadata":"Computational Thinking evaluation"
OR "Full Text & Metadata":"CT evaluation"
OR "Full Text & Metadata":"Computational Thinking framework"
OR "Full Text & Metadata":"CT framework"
OR "Full Text & Metadata":"Computational Thinking assessment"
OR "Full Text & Metadata":"CT assessment")

AND

("Full Text & Metadata":"csl"
OR "Full Text & Metadata":"python course"
OR "Full Text & Metadata":"programming"
OR "Full Text & Metadata":"coding"
OR "Full Text & Metadata":"code")
```

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Initial retrieval: 149 records. After excluding records related to public-key cryptography, convolutional neural networks, and IoT applications, and removing four early records unrelated to educational contexts, 126 records were retained (1990–2025).

(2) CT Intervention / Development Query

```
("Full Text & Metadata":"Computational Thinking intervention"
OR "Full Text & Metadata":"CT intervention"
OR "Full Text & Metadata":"Computational Thinking development"
OR "Full Text & Metadata":"CT development")
```

```
OR "Full Text & Metadata":"Computational Thinking integration"
OR "Full Text & Metadata":"CT integration")
```

AND

```
("Full Text & Metadata":"cs1"
OR "Full Text & Metadata":"python course"
OR "Full Text & Metadata":"programming"
OR "Full Text & Metadata":"coding"
OR "Full Text & Metadata":"code")
```



Initial retrieval: 174 records. Oldest results were not related to educational applications of computational thinking; the search was restricted to 2012 onward, retaining 164 records.

Scopus

Search Field

TITLE-ABS-KEY

Scopus does not provide full-text searching. Searches were therefore conducted using the TITLE-ABS-KEY() function, which searches titles, abstracts, and keywords.

(1) CT Evaluation / Framework Query

```
TITLE-ABS-KEY(
  ("Computational Thinking evaluation"
  OR "CT evaluation"
  OR "Computational Thinking framework"
  OR "CT framework"
  OR "Computational Thinking assessment"
  OR "CT assessment")
AND
  ("cs1" OR "python course" OR "programming"
  OR "coding" OR "code")
)
```



Initial retrieval: 143 records. Records from Medicine and Dentistry were excluded as unrelated to the study objectives. After removal of non-relevant preprints and secondary records, 131 records remained (2001–2025).

(2) CT Intervention / Development Query

```
TITLE-ABS-KEY(
  ("CT intervention"
  OR "Computational Thinking intervention"
  OR "CT development"
  OR "Computational Thinking development"
  OR "CT integration"
  OR "Computational Thinking integration")
```

```
AND
  ("cs1" OR "python course" OR "programming"
   OR "coding" OR "code")
)
```



Retrieval: 156 records. No disciplinary exclusions required. One non-English record was removed, retaining 155 records (2008–2025).

ERIC

Search Field	Title, Abstract, and Keywords
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Initial searches using the full set of evaluation-related terms produced a large number of irrelevant results. The query was therefore simplified while preserving its conceptual intent.

(1) CT Evaluation / Framework Query

```
("Computational Thinking evaluation"
 OR "Computational Thinking framework"
 OR "Computational Thinking assessment")

AND

("cs1" OR "python course" OR "programming"
 OR "coding" OR "code")
```



Original search returned 6,733 records. After query refinement, the final search retrieved 27 records (2006–2025).

(2) CT Intervention / Development Query

```
("CT intervention"
 OR "Computational Thinking intervention"
 OR "CT development"
 OR "Computational Thinking development"
 OR "CT integration"
 OR "Computational Thinking integration")

AND

("cs1" OR "python course" OR "programming"
 OR "coding" OR "code")
```



Final retrieval: 74 records (2006–2025).

ScienceDirect

Search Field

Full text, title, abstract, and author keywords

ScienceDirect imposes restrictions on query length and Boolean complexity. The evaluation query was simplified to remain within the platform's limits while preserving the intended conceptual scope.

(1) Evaluation / Framework Query

```
("Computational Thinking evaluation"  
OR "Computational Thinking framework"  
OR "Computational Thinking assessment")  
  
AND  
  
("cs1" OR "python course" OR "programming"  
OR "coding" OR "code")
```



Final retrieval: 48 records (2017–2025). No disciplinary exclusions were required.

(2) CT Intervention / Development Query

```
("CT intervention"  
OR "Computational Thinking intervention"  
OR "CT development"  
OR "Computational Thinking development"  
OR "CT integration"  
OR "Computational Thinking integration")  
  
AND  
  
("cs1" OR "python course" OR "programming"  
OR "coding" OR "code")
```



Final retrieval: 35 records. No records from unrelated medical or health domains were identified; no additional disciplinary filtering was required.